Computational Fluid Dynamics
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We want to simulate the flow through the duct sketched in Figure 1. The problem is governed by the the steady incompressible Navier-Stokes equations in the domain $\Omega = [0,4] \times [0,1]$:

$$\begin{cases} \rho(\mathbf{u} \cdot \nabla \mathbf{u}) - \operatorname{div} \boldsymbol{\sigma} = \mathbf{0} & \text{in} \quad \Omega, \\ \operatorname{div} \mathbf{u} = \mathbf{0} & \text{in} \quad \Omega, \\ \mathbf{u} = [y(1-y), 0] & \text{on } \Gamma_{\text{in}} = \{(x,y) \in \partial\Omega : x = 0\}, \\ \boldsymbol{\sigma}\mathbf{n} = \mathbf{0}, & \text{on } \Gamma_{\text{out}} = \{(x,y) \in \partial\Omega : x = 5\}, \\ \mathbf{u} = \mathbf{0} & \text{su } \Gamma_{\text{bot}} = \{(x,y) \in \partial\Omega : y = 0\}, \\ (\boldsymbol{\sigma}\mathbf{n}) \cdot \mathbf{t} + \alpha \mathbf{u} \cdot \mathbf{t} = 0; \quad \mathbf{u} \cdot \mathbf{n} = 0, & \text{on } \Gamma_{\text{top}} = \{(x,y) \in \partial\Omega : y = 1\}, \end{cases} \tag{1}$$

where $\boldsymbol{\sigma} = \mu\left(\nabla \mathbf{u} + \nabla \mathbf{u}^T\right) - pI$ is the Cauchy stress tensor, $\mathbf{n}$ and $\mathbf{t} = \mathbf{n}^{\perp}$ are the normal and tangential unit vectors, respectively, and α is a positive constant. The Robin boundary condition on Γ_{top} is a friction condition depending on the constant α.

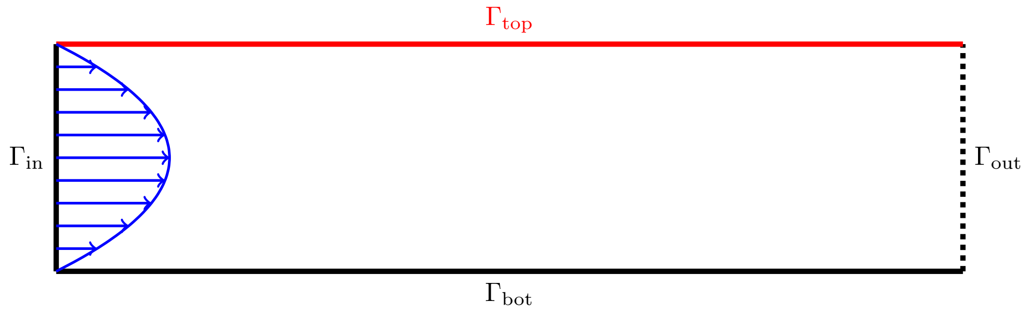

Figure 1: Computational domain

a) Introduce a linearization of Problem (1) based on Newton method.

b) Write the weak formulation of the linear problem to be solved at each Newton iteration.

c) Based on the template `exam.edp`, write a `FreeFem++` script for the solution of Problem (1) using $\mathbb{P}_2 - \mathbb{P}_1$ finite elements for velocity and pressure, respectively, choosing a suitable initial condition and a suitable convergence criterion for the Newton method. Consider the pressure matrix approach for the solution of the system adopting an optimal preconditioner for the Schur complement.

d) Compute the solution considering a fluid with dynamic viscosity $\mu = 0.1$ and density $\rho = 1$. an uniform grid with h=0.2, 0.1, 0.05. Based on the numerical results, discuss the optimality of the preconditioner with respect to the space resolution with α=100.

e) Compute the tangential force acting on the oscillating plate using the grid with h=0.05 and considering different values of $\alpha = 0.01, 1, 100$. For each value of α, compute the tangential force on the top and bottom boundaries and discuss the results.